

Human-Robot Interaction Assignment Report

Simulation scenario:-

In the given simulation a maze is created in which obstacles are included along with the boundaries and a certain path is assigned to the robot which it should follow to reach the final destination without colliding into other objects.

Task assigned:-

The robot must strictly follow the path given to it to reach the goal. The path is zigzag as well as curvy with obstacles as well as it has a specified boundary only in which

Way of approach:-

1. Knowing where I am

My first action is to use my sensors—LiDAR and cameras—to scan the environment around me. I construct a mental model of the world, pinpointing obstacles, barriers, and free space. One can call this step creating a visual map of the world in my brain. Each observation adds to the knowledge of where I can and cannot go.

2. Finding the best route

Once I have learned the layout, I use an algorithm like A* to find the best path to my goal. This allows me to predict and model the most effective and safest route to my destination. I consider all possible paths, considering both distance and any obstacles, and choose the best one.

3. Navigation and Obstacle Avoidance

As I go along my charted course, I am always vigilant. My sensors are continually scanning, and if something—be it a box or a person—gets in my way, I respond quickly. I veer slightly from my course to go around it and then easily return to my original path. I make sure my movements are fluid and efficient.

4. Staying Fast and Efficient

Speed is very important to me. I try as much as possible to travel fast while running with as little chance of collision as possible. I adjust my path to eliminate unnecessary stops or turns. My inbuilt control systems ensure that every move I make is purposeful, precise, and fast.

5. Adapting as I Go

The world doesn't stand still, and neither do I. As soon as something changes—for example, a new obstacle appears, or my goal moves—I immediately update my map and recalculate the best path. Always thinking ahead, I adjust to make sure I reach my destination as quickly and safely as possible.

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Algorithm Overview

Initialization (sysCall_init):

Get object handles for the robot, its references (for orientation, motion), motors, collision volumes, and the goal dummy.

Create a collection of obstacles for collision checking.

Define several parameters for the robot movement, path planning algorithm, and visualisation options.

Cleanup (sysCall_cleanup):

Set the parent of the goal dummy back to the robot when the simulation is terminated.

Collision Checking (checkCollidesAt):

Temporarily move the collision volume of the robot to a specified position.

Check if the collision volume intersects with any obstacles.

Target Position (getTargetPosition):

Introduce a delay to get the target position in order to simulate latency.

Smoothly track and stabilise the goal position.

Path Visualisation (visualizePath):

Draw lines representing the computed path for the robot if available.

Visualisation of Collision-Free Nodes (visualizeCollisionFreeNodes):

Placeholder for the display of intermediate states explored by the planner

Main Thread (sysCall_thread):

Collision-Free Start: Move the robot to a non-colliding configuration if it does not already start that way by repositioning iteratively

Dynamic Goal Adjustment: Adjust the goal if it is unreachable or collides with obstacles to find a valid location

Path Planning:

Set up an OMPL planning instance.

Define the state space (robot's 2D position), collision pairs, and start/goal states.

Run the chosen planning algorithm to find a collision-free path.

Path Execution:

Interpolate along the computed path.

Adjust motor velocities based on the relative position of the target point with respect to the robot.

Dynamically check if the goal has moved and update the path as needed.

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Visualisation and Cleanup:

Visualise the computed path and tracked positions.

Stop the robot's motors and reset visualisations when done.

Motion Control:

Compute motor speeds for a given angle to the next waypoint on the path.

Ensure smooth, bump-free motion.

So in the end by following these steps I was able to successfully route the robot from starting position to ending position.